TRAINER PREPARATION GUIDE

Linux Administration Essentials

Instructor Copy

LNX-001

1-day course · 4 modules · Beginner level · Bootcamp Mode

Instructor Copy

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Introduction

This Trainer Preparation Guide is designed to help you prepare for and deliver **Linux Administration Essentials** effectively. It is an instructor-only document and should not be distributed to students.

Use this guide before class to review timing recommendations, refresh your knowledge of module Q&As, and decide whether to administer the pre and post assessments. During class, the consolidated Q&A reference and timing table provide quick access to the information you need most.

> **Assessments are optional.** The pre and post assessment tests at the end of this guide are provided at your discretion. Student question-only handouts are formatted to print cleanly from any section. Answer keys immediately follow each student handout.

Delivery Options

You have several ways to use the assessments, Q&A reference, and Quick Reference Guides in this guide. Choose what fits your class.

- **Pre-assessment:** administer at course start as a baseline, at the start of each module as a formative check, or skip entirely.
- **Post-assessment:** administer at course end as a summative check, at the end of each module as a formative checkpoint, or both.
- **Same or different items:** use the same items pre and post to measure growth, or use the parallel items provided to prevent memorization-only learning.
- **Grading:** grade for completion, grade for score, treat as ungraded baseline, or have students self-score.
- **Module Q&A:** use as in-class discussion prompts, take-home review, full answer-key handout, or verbal walkthrough.
- **Quick Reference Guides:** hand out as a pre-class preview to anchor the day, as an end-of-day takeaway reference, or on demand during the class.

Lab Environment Setup

Students will use a local computer or virtual machine running Ubuntu 24 on which they have root access. Any scripts or resources needed for labs must be downloadable from the Internet or created by them during the lab exercise. Instructions for installing tools not native to Ubuntu 24 must be provided in the lab instructions.

Trainer Prerequisites

Before teaching this course, you should have the following skills and experience:

- Solid working knowledge of the subject matter covered in all 4 modules
- Basic familiarity with general computing concepts and operating systems, with no prior Linux command-line experience required.
- Experience facilitating instructor-led technical training (ILT)
- Familiarity with the lab environment and any required platforms or tools
- Experience running intensive Bootcamp-style sessions with capstone labs

Course Learning Objectives

Students will be able to:

- Navigate the Ubuntu filesystem and execute essential administrative commands through the terminal.
- Manage local user accounts, groups, and file permissions using standard Linux administrative utilities.
- Inspect system hardware, monitor running processes, and manage local storage filesystems.
- Configure network interfaces and verify local and remote network connectivity on Ubuntu hosts.
- Implement foundational host security using SSH configurations, sudo privileges, and Uncomplicated Firewall (UFW) rules.

Recommended Timings

Use this section to plan pacing before class. Scheduled time reflects the approved syllabus. Module totals include lecture/demo time, required lab time, and an adaptive 5-30 minute review/summary block based on module size and review-question count. Lunch and breaks are not included in module totals.

Module	Title	Scheduled Time	Lecture & Demo	Required Lab	Review/Summary	Required Total
1	Linux CLI Foundations and Filesystem Navigation	**80 min**	40 min	30 min	10 min	**80 min**
2	User, Group, and Permission Administration	**90 min**	45 min	30 min	15 min	**90 min**
3	Hardware Detection, Storage, and Process Management	**95 min**	45 min	35 min	15 min	**95 min**
4	Host Networking and Baseline System Security	**95 min**	45 min	35 min	15 min	**95 min**
	Day 1 — Capstone Lab: Edge Sensor Station Provisioning and Security Hardening	—	—	240 min	—	**240 min**
Total		**360 min (6h 0m)**	**175 min**	**130 min**	**55 min**	**360 min**

Lunch and breaks are scheduled by the instructor, training center, or client according to local policy. A typical training day includes a 30–60 minute lunch and 2–4 short breaks of 10–15 minutes. Breaks are commonly placed every 45–60 minutes based on class pace, lab intensity, and facility requirements.

Review/summary blocks may be delivered live, shortened, combined, or assigned as self-study with answers provided later. Required labs should be protected whenever possible.

Bootcamp Mode Preparation

This course was built with **Bootcamp Mode** enabled. **Bootcamp Mode is an optional alternative to standard Instructor-Led Training (ILT) — not an addition to it.** When Bootcamp Mode is used, mornings are devoted to the step-by-step module labs and afternoons are devoted to scenario-based Capstone Labs that integrate the day's skills. The instructor does not deliver the lecture portions of the modules in this format. ILT and Bootcamp Mode are mutually exclusive — choose one delivery method for the class. A standalone Capstone Slides deck (CapstoneSlides.pptx) is included in the deliverable bundle and may be used at the instructor's discretion when running this course in Bootcamp Mode.

Key Differences from Standard ILT

- Each day ends with a Capstone Lab — a complex, scenario-driven exercise that requires students to synthesize skills from multiple modules
- Capstone Labs are time-intensive. Budget 60–90 minutes per capstone, including debrief
- Maintain a faster pace through module content to preserve time for capstone work
- Use discussion prompts to transition between modules and keep energy high

Capstone Facilitation Tips

> **Before the Capstone:** Briefly recap the day's modules and explain how the capstone integrates those skills. Set clear expectations on deliverables and time limits.
>
> **During the Capstone:** Circulate and check in with students. Note common blockers and be ready to offer hints without giving away solutions. Encourage peer collaboration where appropriate.
>
> **After the Capstone:** Conduct a class debrief — ask 2–3 students to walk through their approach. Highlight different solutions and common mistakes. Connect outcomes back to real-world application.

Capstone Lab Schedule

The table below lists the Capstone Labs generated for this course. Each capstone occupies the afternoon block of its respective day and integrates the day's combined skills.

Day	Capstone Title	Source Modules	Skills Integrated	Est. Minutes
1	Edge Sensor Station Provisioning and Security Hardening	1, 2, 3, 4	• Filesystem hierarchy navigation, file manipulation, and standard stream redirection pipelines • Local user account provisioning, supplementary group management, and POSIX permission boundary enforcement • Block device partitioning with fdisk, ext4 formatting, persistent UUID mounting in /etc/fstab, and process execution control • Netplan static network configuration, OpenSSH daemon access restriction, selective sudo delegation, and UFW packet filtering	240 min

Module Q&A Reference

All end-of-module review questions and answers are consolidated here for quick reference during class. Use these during module review sessions, Q&A breaks, or to prepare for anticipated student questions.

Module 1: Linux CLI Foundations and Filesystem Navigation

1. Which of the following represents an absolute path in the Linux Filesystem Hierarchy Standard?

A) etc/netplan/01-netcfg.yaml

B) ../var/log/syslog

C) /var/log/auth.log

D) ./scripts/backup.sh

> **Answer: C**
>
> *An absolute path always begins with the filesystem root directory, denoted by an initial forward slash (/). Paths beginning with directory names, single dots (.), or double dots (..) are relative paths.*

2. Which standard FHS directory stores system-wide host configuration files and administrative settings?

A) /home

B) /etc

C) /usr/bin

D) /var

> **Answer: B**
>
> *The /etc directory contains host-specific system configuration files. /home hosts personal user files, /var contains variable data like logs, and /usr/bin stores executable application binaries.*

3. An administrator wants to write command output to a file named audit.log without deleting existing log entries. Which operator must they use?

A) >

B) <

C) |

D) >>

> **Answer: D**
>
> *The >> operator appends standard output to the end of the specified target file without overwriting existing data. The single > operator truncates and overwrites the target file.*

4. What is the primary function of the Linux pipeline operator (|)?

A) It directs the standard output (stdout) of one command into the standard input (stdin) of another command.

B) It runs two independent commands at the same time in the background.

C) It saves standard error (stderr) directly into an existing disk file.

D) It pauses execution until the user presses the Enter key.

> **Answer: A**
>
> *The pipeline operator (|) connects the standard output stream of the preceding command directly into the standard input stream of the succeeding command, allowing data filtering and chaining without temporary files.*

Module 2: User, Group, and Permission Administration

1. Which command correctly appends the secondary group 'devops' to an existing user named 'jdoe' without removing existing group memberships?

A) sudo usermod -g devops jdoe

B) sudo usermod -aG devops jdoe

C) sudo groupadd -u jdoe devops

D) sudo useradd -G devops jdoe

> **Answer: B**
>
> *The `-aG` options in `usermod` append (`-a`) the user to the specified secondary group (`-G`) without stripping memberships from other existing supplementary groups.*

2. Where are salted, encrypted user password hashes and account expiration parameters stored in a standard Linux system?

A) /etc/passwd

B) /etc/group

C) /etc/shadow

D) /etc/security/limits.conf

Answer: C

The `/etc/shadow` file holds sensitive password hashes and account lifecycle aging parameters and is readable only by administrative accounts. `/etc/passwd` holds general account metadata.

3. What permission does a user require on a parent directory in order to change their current working directory into it?

A) Read (r)

B) Write (w)

C) Execute (x)

D) Modify (m)

Answer: C

On directories, the execute (`x`) permission enables search and traversal, permitting users to change into the directory with `cd` and access nested files.

4. What are the effective permissions on a regular file if an administrator executes 'chmod 640 secret.txt'?

A) User: rw-, Group: r--, Other: ---

B) User: rwx, Group: r-x, Other: ---

C) User: rw-, Group: rw-, Other: r--

D) User: r--, Group: r--, Other: ---

Answer: A

Octal 6 equals 4+2 (read and write, `rw-`), 4 equals read-only (`r--`), and 0 equals no permissions (`---`). Therefore, 640 translates to `rw-r-----`.

5. Which command simultaneously updates both the owning user to 'webadmin' and the owning group to 'www-data' on a directory named '/var/www/html'?

A) sudo chmod webadmin:www-data /var/www/html

B) sudo chown webadmin:www-data /var/www/html

C) sudo chgrp webadmin:www-data /var/www/html

D) sudo usermod -u webadmin -g www-data /var/www/html

Answer: B

The `chown` command supports the `OWNER:GROUP` syntax, allowing an administrator to update both user ownership and group ownership in a single operation.

Module 3: Hardware Detection, Storage, and Process Management

1. Which command displays a hierarchical tree structure of all recognized block storage devices and their associated mount points?

A) lspci

B) lsusb

C) lsblk

D) lshw -short

Answer: C

The lsblk command lists block devices, including hard drives, partitions, and virtual disks, in a tree-like format showing device names, sizes, and active mount points. lspci inspects PCI devices, lsusb lists USB controllers, and lshw gives overall hardware summaries.

2. What is the recommended administrative command to format an existing partition (/dev/sdb1) with an ext4 filesystem?

A) sudo fdisk /dev/sdb1

B) sudo mkfs.ext4 /dev/sdb1

C) sudo mount -t ext4 /dev/sdb1

D) sudo blkid -f ext4 /dev/sdb1

Answer: B

mkfs.ext4 builds an ext4 filesystem on a specified partition. fdisk is used to manage partition tables, mount attaches already-formatted filesystems to directories, and blkid discovers filesystem attributes and UUIDs.

3. Why should a system administrator issue SIGTERM (signal 15) to an unresponsive process before resorting to SIGKILL (signal 9)?

A) SIGTERM requires administrative root privileges, whereas SIGKILL can be executed by any user.

B) SIGKILL cannot be run against processes that have children or thread pools.

C) SIGTERM allows the target process to release memory locks, save state, and perform cleanup routines before exiting.

D) SIGKILL pauses the process in memory rather than removing it from the kernel task list.

Answer: C

SIGTERM (15) requests termination while allowing the process to handle the signal, close open files, and perform clean exits. SIGKILL (9) halts the process immediately at the kernel level without cleanup, which can leave corrupted data and lock files behind.

4. When attempting to run 'sudo umount /mnt/data', the command fails with the error message 'target is busy'. What is the most common cause of this error?

A) The filesystem was formatted with an incompatible ext4 block size.

B) The current working shell or an active background process is inside or accessing /mnt/data.

C) The partition lacks a defined UUID in /etc/fstab.

D) The mount command was executed with the read-only flag.

Answer: B

A 'target is busy' error occurs when the kernel detects that an open file descriptor, running process, or active shell working directory is currently located within the mount point hierarchy. Moving out of the directory (e.g., cd ~) resolves the issue.

5. Which command tests and mounts all filesystems specified in /etc/fstab without requiring a system reboot?

A) sudo mount -a

B) sudo fdisk -l

C) sudo blkid --refresh

D) sudo systemctl reload fstab

Answer: A

The 'mount -a' command reads /etc/fstab and attempts to mount all filesystems marked with default mount configurations that are not already mounted. It is the primary safety check used to verify /etc/fstab syntax before rebooting.

Module 4: Host Networking and Baseline System Security

1. Which modern command displays all network interfaces, their assigned IP addresses, and their current link state on Ubuntu 24?

A) ifconfig -a

B) ip addr show

C) netstat -in

D) route -e

Answer: B

The `ip addr show` (or `ip a`) command from the modern `iproute2` suite displays interfaces, IP addresses, and operational link states. The legacy tool `ifconfig` is deprecated.

2. What is the primary formatting requirement when configuring Netplan YAML files in /etc/netplan/?

A) Indentation must use space characters rather than tab characters

B) Lines must terminate with semicolons

C) Section headers must be enclosed in square brackets

D) Values must always be delimited with single quotes

Answer: A

Netplan uses YAML syntax, which strictly forbids tab characters for indentation and requires spaces. Using tabs causes parsing failures when running `netplan apply`.

3. Which command displays active TCP and UDP listening sockets along with their process names and numeric port numbers?

A) ping -tulpn

B) traceroute -l

C) ss -tulpn

D) ip link show -listening

Answer: C

The `ss -tulpn` command displays TCP (`-t`), UDP (`-u`), listening (`-l`), process names (`-p`), and numeric addresses/ports (`-n`).

4. What critical step should an administrator perform before running 'sudo ufw enable' on a remote Ubuntu server?

A) Change the default root password to blank

B) Explicitly allow SSH connections using 'sudo ufw allow OpenSSH'

C) Uninstall the OpenSSH server package

D) Reboot the system into single-user rescue mode

Answer: B

Because UFW defaults to denying all inbound connections, enabling the firewall without explicitly allowing OpenSSH (`sudo ufw allow OpenSSH` or `sudo ufw allow 22/tcp`) will immediately lock out remote administrators.

5. Why must administrators use the 'visudo' command rather than a standard text editor to modify sudo privilege files?

A) visudo automatically reboots the system to apply new group policies

B) visudo enforces file locking and performs syntax validation to prevent administrators from being locked out

C) Standard text editors like nano cannot open files located in /etc

D) visudo encrypts the sudoers file with an asymmetric hardware key

Answer: B

`visudo` provides file locking to avoid concurrent edit conflicts and validates the syntax of the sudoers file prior to saving. Syntax errors in `/etc/sudoers` can completely disable `sudo` functionality across the system.

Classroom Handouts

The materials in this section are designed to be handed to students at your discretion. Use them in the way that best fits your class — see the Delivery Options earlier in this guide for ideas.

Quick Reference Guides

One single-page Quick Reference Guide per training day. Each guide distills the day's most useful on-the-job tasks — designed for students to keep and reference at work after class.

Quick Reference Guide

Day 1 — Linux Administration Essentials

LNX-001

Day 1 · Modules 1, 2, 3, 4

How to use this guide: Essential on-the-job command reference for Ubuntu 24 system administration.

Task 1 — You need to search and filter system logs using pipelines and redirection

Locate entries: Filter log files for case-insensitive keywords using grep piped to pagers or summary counts. `grep -i` | `wc -l`

Direct stream output: Use single bracket redirection to overwrite target report files or double brackets to append updates. `>` `>>`

Follow live logs: Monitor incoming events in real time during troubleshooting sessions with the follow flag. `tail -f /var/log/syslog`

```
grep -i "error" /var/log/syslog | tail -n 20
grep "/bin/bash" /etc/passwd | wc -l > /tmp/bash_users_count.txt
```

Task 2 — You need to provision a new user account and assign supplementary group access

Create user identity: Provision the account with a home directory and interactive login shell in a single step. `useradd -m -s /bin/bash <user>`

Set credentials: Assign an initial password to the account to enable standard user authentication. `sudo passwd <user>`

Preserve memberships: Always use append mode when assigning secondary groups to prevent stripping existing group permissions. `usermod -aG <group> <user>`

Verify identity: Confirm assigned numerical identifiers and complete group memberships before releasing access. `id <user>`

```
sudo useradd -m -s /bin/bash arivera
sudo passwd arivera
sudo usermod -aG developers arivera
id arivera
```

Task 3 — You need to secure a shared directory with group ownership and boundary permissions

Assign directory ownership: Update both the owning user and owning group simultaneously using standard colon syntax. `chown <user>:<group> <path>`

Enforce group boundaries: Set octal mode 770 so group members have full access while unprivileged others are denied. `chmod 770 <path>`

Verify applied modes: Inspect target directory metadata to verify security tiers before placing collaborative files inside. `ls -ld <path>`

```
sudo chown -R arivera:developers /srv/projects
sudo chmod 770 /srv/projects
ls -ld /srv/projects
```

Octal Mode	POSIX String	Effective Access Rights
770	rwxrwx---	Full access for owner and group; no access for others
750	rwxr-x---	Full owner access, group read/execute, no access for others
640	rw-r-----	Owner read/write, group read-only, no access for others
644	rw-r--r--	Owner read/write, world read-only

Task 4 — You need to format a storage partition and configure persistent mounting in /etc/fstab

Format partition: Create an ext4 filesystem on the target block device partition using the standard filesystem builder. `mkfs.ext4 /dev/sdb1`

Retrieve filesystem UUID: Obtain the unique identifier to ensure persistent mounting independent of drive letter shifts. `sudo blkid /dev/sdb1`

Define fstab entry: Append the UUID definition specifying mount point, filesystem type, defaults options, dump, and fsck pass. `/etc/fstab`

Validate before reboot: Test mounting all configured filesystems immediately to catch syntax errors before the host reboots. `sudo mount -a`

```
sudo mkfs.ext4 /dev/sdb1
sudo blkid /dev/sdb1
# Example /etc/fstab entry: UUID=<UUID_VALUE> /mnt/data ext4 defaults 0 2
sudo mount -a
```

Task 5 — You need to locate and terminate an unresponsive background process safely

Locate process identifier: Find the active process ID using snapshot listing or pattern-matching tools. `ps aux | grep <name>` `pgrep -fl <name>`

Attempt graceful stop: Send termination signal 15 first so the application can flush buffers and close open locks cleanly. `kill -15 <PID>`

Force kill if stuck: Issue signal 9 only when the process ignores graceful termination and remains unresponsive. `kill -9 <PID>`

```
pgrep -fl worker_script
kill -15 <PID>
# If process fails to terminate after several seconds:
kill -9 <PID>
```

Task 6 — You need to configure host firewall rules with UFW without losing remote SSH access

Set default policies: Enforce baseline host defense by blocking all unsolicited incoming packets while allowing outbound connections. `ufw default deny incoming` `ufw default allow outgoing`

Allow SSH management: Explicitly permit OpenSSH connections before activating the firewall to prevent immediate administrative lockout. `sudo ufw allow OpenSSH`

Enable and verify: Turn on firewall packet filtering and review active rule tables in verbose mode. `sudo ufw enable` `sudo ufw status verbose`

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow OpenSSH
sudo ufw enable
sudo ufw status verbose
```

LMS-Embedded AI Learning Assistant (LMS Packages)

If your course package included the **LMS-Embedded AI Learning Assistant** in its LMS files (SCORM, xAPI, or cmi5 packages), learners can ask questions and receive course-grounded answers directly inside the e-learning modules. The LMS-Embedded AI Learning Assistant is not part of the Word, PowerPoint, or PDF deliverables — it applies only to the LMS package formats.

Enabling or disabling the LMS-Embedded AI Learning Assistant. The LMS-Embedded AI Learning Assistant is controlled by a single on/off switch inside the LMS package, in the file ai-tutor/tutor-config.js. Open that file and change the value on its last line:

- window.__CS_AI_TUTOR_ENABLED__ = 1; — LMS-Embedded AI Learning Assistant **enabled** (default)
- window.__CS_AI_TUTOR_ENABLED__ = 0; — LMS-Embedded AI Learning Assistant **disabled**

Change **only** that number — do not rename, move, or delete any files in the ai-tutor/ folder, as the package manifest expects all of them to be present. Setting the value to 0 fully disables the Tutor while leaving every file in place.

Important: edit this file *before* importing the package into your LMS. Once a package is imported, the LMS keeps its own copy, so editing the original .zip afterward has no effect. To change the setting after import, edit the file, re-zip the package, and re-import it.

Pre-Assessment — Student Handout

Questions: 10 **Time Limit:** 20 minutes **Passing Score:** 7 / 10

Instructions: Answer each question to the best of your ability. Circle or write the letter of the best answer. This is a prerequisite knowledge check — do your best even if you have not studied this material yet.

1. What permission does a user require on a parent directory in order to change their current working directory into it?

A. Read (r)
B. Write (w)
C. Execute (x)
D. Modify (m)

2. Which command simultaneously updates both the owning user to 'webadmin' and the owning group to 'www-data' on a directory named '/var/www/html'?

A. sudo chmod webadmin:www-data /var/www/html
B. sudo chown webadmin:www-data /var/www/html
C. sudo chgrp webadmin:www-data /var/www/html
D. sudo usermod -u webadmin -g www-data /var/www/html

3. Why should a system administrator issue SIGTERM (signal 15) to an unresponsive process before resorting to SIGKILL (signal 9)?

A. SIGTERM requires administrative root privileges, whereas SIGKILL can be executed by any user.
B. SIGKILL cannot be run against processes that have children or thread pools.
C. SIGTERM allows the target process to release memory locks, save state, and perform cleanup routines before exiting.
D. SIGKILL pauses the process in memory rather than removing it from the kernel task list.

4. What distinguishes an absolute filesystem path from a relative path on an Ubuntu system?

A. An absolute path must begin with the root directory slash (/), while a relative path begins from the current working directory without a leading slash.

B. An absolute path refers only to system configuration directories under /etc, while a relative path references user home folders.

C. An absolute path requires dot-slash (./) prefixing, whereas a relative path starts with a forward slash.

D. An absolute path can only be executed by the root user, while relative paths are unprivileged.

5. Which Linux core utility is designed specifically to create empty files or update file timestamps?

A. mkdir

B. touch

C. cp

D. mv

6. In Linux standard stream architecture, which file descriptor number represents standard error (stderr)?

A. 0

B. 1

C. 2

D. 3

7. In the Linux user identity model, which numerical User Identifier (UID) is permanently reserved for the root superuser?

A. UID 0

B. UID 1

C. UID 100

D. UID 1000

8. Which configuration file in /etc stores encrypted user password hashes and account expiration parameters?

A. /etc/passwd
B. /etc/shadow
C. /etc/group
D. /etc/login.defs

9. When using chmod in symbolic mode, which character represents the file's owning group class?

A. u
B. g
C. o
D. a

10. Which administrative utility is used to inspect peripheral components connected via Peripheral Component Interconnect (PCI) buses?

A. lsusb
B. lspci
C. lsblk
D. fdisk

Pre-Assessment — Instructor Answer Key

Instructor Note: Whether to administer this assessment is entirely at your discretion. The student handout on the preceding pages can be printed and distributed before or after the relevant course content. Answers may be reviewed as a class discussion, distributed as a separate handout, or withheld. Use your judgment based on the experience level of your students and the goals of the session.

Passing Score: 7 of 10 correct **Time Limit:** 20 minutes

1. What permission does a user require on a parent directory in order to change their current working directory into it?

A. Read (r)

B. Write (w)

C. Execute (x)

D. Modify (m)

Answer: C

On directories, the execute (`x`) permission enables search and traversal, permitting users to change into the directory with `cd` and access nested files.

2. Which command simultaneously updates both the owning user to 'webadmin' and the owning group to 'www-data' on a directory named '/var/www/html'?

A. sudo chmod webadmin:www-data /var/www/html

B. sudo chown webadmin:www-data /var/www/html

C. sudo chgrp webadmin:www-data /var/www/html

D. sudo usermod -u webadmin -g www-data /var/www/html

Answer: B

The `chown` command supports the `OWNER:GROUP` syntax, allowing an administrator to update both user ownership and group ownership in a single operation.

3. Why should a system administrator issue SIGTERM (signal 15) to an unresponsive process before resorting to SIGKILL (signal 9)?

A. SIGTERM requires administrative root privileges, whereas SIGKILL can be executed by any user.

B. SIGKILL cannot be run against processes that have children or thread pools.

C. SIGTERM allows the target process to release memory locks, save state, and perform cleanup routines before exiting.

D. SIGKILL pauses the process in memory rather than removing it from the kernel task list.

Answer: C

SIGTERM (15) requests termination while allowing the process to handle the signal, close open files, and perform clean exits. SIGKILL (9) halts the process immediately at the kernel level without cleanup, which can leave corrupted data and lock files behind.

4. What distinguishes an absolute filesystem path from a relative path on an Ubuntu system?

A. An absolute path must begin with the root directory slash (/), while a relative path begins from the current working directory without a leading slash.

B. An absolute path refers only to system configuration directories under /etc, while a relative path references user home folders.

C. An absolute path requires dot-slash (./) prefixing, whereas a relative path starts with a forward slash.

D. An absolute path can only be executed by the root user, while relative paths are unprivileged.

Answer: A

Absolute paths always begin from the root directory (/), whereas relative paths start from the current working directory without a leading slash.

5. Which Linux core utility is designed specifically to create empty files or update file timestamps?

A. mkdir

B. touch

C. cp

D. mv

Answer: B

The touch command is used to update file access timestamps or create a blank file if the target file does not already exist.

6. In Linux standard stream architecture, which file descriptor number represents standard error (stderr)?

A. 0

B. 1

C. 2

D. 3

Answer: C

Standard streams are assigned numeric file descriptors: standard input (stdin) is 0, standard output (stdout) is 1, and standard error (stderr) is 2.

7. In the Linux user identity model, which numerical User Identifier (UID) is permanently reserved for the root superuser?

A. UID 0

B. UID 1

C. UID 100

D. UID 1000

Answer: A

The Linux kernel identifies the root superuser account with UID 0, granting it unrestricted administrative privileges.

8. Which configuration file in /etc stores encrypted user password hashes and account expiration parameters?

A. /etc/passwd

B. /etc/shadow

C. /etc/group

D. /etc/login.defs

Answer: B

The /etc/shadow file contains encrypted password hashes and account lifecycle aging parameters, readable only by administrative accounts.

9. When using chmod in symbolic mode, which character represents the file's owning group class?

A. u

B. g

C. o

D. a

Answer: B

In symbolic chmod syntax, 'g' specifies the group ownership class ('u' is user/owner, 'o' is others, and 'a' is all).

10. Which administrative utility is used to inspect peripheral components connected via Peripheral Component Interconnect (PCI) buses?

A. lsusb

B. lspci

C. lsblk

D. fdisk

Answer: B

The lspci utility lists and inspects devices connected to PCI buses, such as network adapters and storage controllers.

Post-Assessment — Student Handout

Questions: 20 **Time Limit:** 30 minutes **Passing Score:** 16 / 20

Instructions: Answer each question to the best of your ability. Circle or write the letter of the best answer. This is a knowledge retention check — demonstrate what you learned during this course.

1. Which of the following represents an absolute path in the Linux Filesystem Hierarchy Standard?

A. etc/netplan/01-netcfg.yaml

B. ../var/log/syslog

C. /var/log/auth.log

D. ./scripts/backup.sh

2. Why must administrators use the 'visudo' command rather than a standard text editor to modify sudo privilege files?

A. visudo automatically reboots the system to apply new group policies

B. visudo enforces file locking and performs syntax validation to prevent administrators from being locked out

C. Standard text editors like nano cannot open files located in /etc

D. visudo encrypts the sudoers file with an asymmetric hardware key

3. Which standard FHS directory stores system-wide host configuration files and administrative settings?

A. /home

B. /etc

C. /usr/bin

D. /var

4. Where are salted, encrypted user password hashes and account expiration parameters stored in a standard Linux system?

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C. /etc/shadow

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5. What is the primary function of the Linux pipeline operator (|)?

A. It directs the standard output (stdout) of one command into the standard input (stdin) of another command.

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6. An administrator wants to write command output to a file named audit.log without deleting existing log entries. Which operator must they use?

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C. |

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7. What are the effective permissions on a regular file if an administrator executes 'chmod 640 secret.txt'?

A. User: rw-, Group: r--, Other: ---

B. User: rwx, Group: r-x, Other: ---

C. User: rw-, Group: rw-, Other: r--

D. User: r--, Group: r--, Other: ---

8. What critical step should an administrator perform before running 'sudo ufw enable' on a remote Ubuntu server?

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B. Explicitly allow SSH connections using 'sudo ufw allow OpenSSH'

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9. Which modern command displays all network interfaces, their assigned IP addresses, and their current link state on Ubuntu 24?

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10. Which command displays active TCP and UDP listening sockets along with their process names and numeric port numbers?

A. ping -tulpn
B. traceroute -l
C. ss -tulpn
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11. An administrator needs to duplicate an entire directory tree located at /etc/netplan to /tmp/netplan_backup. Which command achieves this?

A. cp /etc/netplan /tmp/netplan_backup
B. cp -r /etc/netplan /tmp/netplan_backup
C. mv -r /etc/netplan /tmp/netplan_backup
D. touch -r /etc/netplan /tmp/netplan_backup

12. A technician starts in `/var` and executes `cd log` followed by `cd ~`. What does `pwd` return immediately after these steps?

A. /var/log
B. /root
C. /home/<username>
D. /

13. An administrator in a staging directory runs `touch inventory_draft.txt` and then wishes to move and rename it to `reports/system_inventory.txt`. Which core command executes this operation in place?

A. cp inventory_draft.txt reports/system_inventory.txt

B. mv inventory_draft.txt reports/system_inventory.txt

C. rm inventory_draft.txt reports/system_inventory.txt

D. touch reports/system_inventory.txt inventory_draft.txt

14. To inspect exactly the first 5 entries of /etc/passwd without opening an interactive editor, which command should you run?

A. head -n 5 /etc/passwd

B. tail -n 5 /etc/passwd

C. cat -n 5 /etc/passwd

D. less -n 5 /etc/passwd

15. An administrator wants to extract lines containing '/bin/bash' from /etc/passwd into bash_users.txt, and then append an audit comment to the end of that same file. Which sequence of operators accomplishes this?

A. Use `>` to extract the lines, then use `>>` to add the comment.

B. Use `>>` to extract the lines, then use `>` to add the comment.

C. Use `|` to extract the lines, then use `>` to add the comment.

D. Use `<` to extract the lines, then use `|` to add the comment.

16. Which command pipeline counts the total number of user accounts assigned the '/usr/sbin/nologin' shell in /etc/passwd?

A. grep "/usr/sbin/nologin" /etc/passwd > wc -l

B. grep "/usr/sbin/nologin" /etc/passwd | wc -l

C. cat /etc/passwd | grep -c | wc -l

D. wc -l < grep "/usr/sbin/nologin" /etc/passwd

17. While appending summary records to an audit log, a junior administrator accidentally runs `echo "Demarcation" > audit.log` instead of using `>>`. What is the result, and how should it be avoided?

A. The command errors out because `>` requires an existing file descriptor; no data is lost.

B. The existing contents of audit.log are truncated and overwritten; `>>` must be used to preserve existing log entries.

C. The echo string is appended to the beginning of the file, shifting previous records downward.

D. The operating system creates a backup copy of audit.log automatically before truncating.

18. While viewing configuration documentation in a terminal, the terminal window enters a `less` pager session and will not accept normal command prompts. How should the administrator return to Bash?

A. Press `Ctrl + C` repeatedly until the terminal aborts.

B. Type `exit` and press Enter.

C. Press `q` on the keyboard.

D. Close the terminal emulator window and start a new session.

19. An administrator provisions a new local account named `developer1`. Which command ensures that the user's home directory is created and default skeleton files are copied?

A. sudo useradd developer1

B. sudo useradd -m developer1

C. sudo usermod -m developer1

D. sudo groupadd -m developer1

20. A user has read (`r`) and write (`w`) permissions on a directory, but lacks execute (`x`) permission. What action is the user prevented from performing?

A. Listing directory contents with ls

B. Changing their current working directory into that directory with cd

C. Viewing file metadata such as timestamps

D. Deleting files that belong to other users

Post-Assessment — Instructor Answer Key

Instructor Note: Whether to administer this assessment is entirely at your discretion. The student handout on the preceding pages can be printed and distributed before or after the relevant course content. Answers may be reviewed as a class discussion, distributed as a separate handout, or withheld. Use your judgment based on the experience level of your students and the goals of the session.

Passing Score: 16 of 20 correct **Time Limit:** 30 minutes

1. Which of the following represents an absolute path in the Linux Filesystem Hierarchy Standard?

A. etc/netplan/01-netcfg.yaml

B. ../var/log/syslog

C. /var/log/auth.log

D. ./scripts/backup.sh

Answer: C

An absolute path always begins with the filesystem root directory, denoted by an initial forward slash (/). Paths beginning with directory names, single dots (.), or double dots (..) are relative paths.

2. Why must administrators use the 'visudo' command rather than a standard text editor to modify sudo privilege files?

A. visudo automatically reboots the system to apply new group policies

B. visudo enforces file locking and performs syntax validation to prevent administrators from being locked out

C. Standard text editors like nano cannot open files located in /etc

D. visudo encrypts the sudoers file with an asymmetric hardware key

Answer: B

`visudo` provides file locking to avoid concurrent edit conflicts and validates the syntax of the sudoers file prior to saving. Syntax errors in `/etc/sudoers` can completely disable `sudo` functionality across the system.

3. Which standard FHS directory stores system-wide host configuration files and administrative settings?

A. /home

B. /etc

C. /usr/bin

D. /var

Answer: B

The /etc directory contains host-specific system configuration files. /home hosts personal user files, /var contains variable data like logs, and /usr/bin stores executable application binaries.

4. Where are salted, encrypted user password hashes and account expiration parameters stored in a standard Linux system?

A. /etc/passwd

B. /etc/group

C. /etc/shadow

D. /etc/security/limits.conf

Answer: C

The `/etc/shadow` file holds sensitive password hashes and account lifecycle aging parameters and is readable only by administrative accounts. `/etc/passwd` holds general account metadata.

5. What is the primary function of the Linux pipeline operator (|)?

A. It directs the standard output (stdout) of one command into the standard input (stdin) of another command.

B. It runs two independent commands at the same time in the background.

C. It saves standard error (stderr) directly into an existing disk file.

D. It pauses execution until the user presses the Enter key.

Answer: A

The pipeline operator (|) connects the standard output stream of the preceding command directly into the standard input stream of the succeeding command, allowing data filtering and chaining without temporary files.

6. An administrator wants to write command output to a file named audit.log without deleting existing log entries. Which operator must they use?

A. >

B. <

C. |

D. >>

> **Answer: D**
>
> *The >> operator appends standard output to the end of the specified target file without overwriting existing data. The single > operator truncates and overwrites the target file.*

7. What are the effective permissions on a regular file if an administrator executes 'chmod 640 secret.txt'?

A. User: rw-, Group: r--, Other: ---

B. User: rwx, Group: r-x, Other: ---

C. User: rw-, Group: rw-, Other: r--

D. User: r--, Group: r--, Other: ---

> **Answer: A**
>
> *Octal 6 equals 4+2 (read and write, `rw-`), 4 equals read-only (`r--`), and 0 equals no permissions (`---`). Therefore, 640 translates to `rw-r-----`.*

8. What critical step should an administrator perform before running 'sudo ufw enable' on a remote Ubuntu server?

A. Change the default root password to blank

B. Explicitly allow SSH connections using 'sudo ufw allow OpenSSH'

C. Uninstall the OpenSSH server package

D. Reboot the system into single-user rescue mode

> **Answer: B**
>
> *Because UFW defaults to denying all inbound connections, enabling the firewall without explicitly allowing OpenSSH (`sudo ufw allow OpenSSH` or `sudo ufw allow 22/tcp`) will immediately lock out remote administrators.*

9. Which modern command displays all network interfaces, their assigned IP addresses, and their current link state on Ubuntu 24?

A. ifconfig -a

B. ip addr show

C. netstat -in

D. route -e

Answer: B

The `ip addr show` (or `ip a`) command from the modern `iproute2` suite displays interfaces, IP addresses, and operational link states. The legacy tool `ifconfig` is deprecated.

10. Which command displays active TCP and UDP listening sockets along with their process names and numeric port numbers?

A. ping -tulpn

B. traceroute -l

C. ss -tulpn

D. ip link show -listening

Answer: C

The `ss -tulpn` command displays TCP (`-t`), UDP (`-u`), listening (`-l`), process names (`-p`), and numeric addresses/ports (`-n`).

11. An administrator needs to duplicate an entire directory tree located at /etc/netplan to /tmp/netplan_backup. Which command achieves this?

A. cp /etc/netplan /tmp/netplan_backup

B. cp -r /etc/netplan /tmp/netplan_backup

C. mv -r /etc/netplan /tmp/netplan_backup

D. touch -r /etc/netplan /tmp/netplan_backup

Answer: B

The cp command requires the -r (recursive) option to copy a directory and its complete nested contents.

12. A technician starts in `/var` and executes `cd log` followed by `cd ~`. What does `pwd` return immediately after these steps?

A. /var/log

B. /root

C. /home/<username>

D. /

> **Answer: C**
>
> *The tilde shortcut (`~`) expands directly to the active user's personal home directory, returning `/home/<username>`.*

13. An administrator in a staging directory runs `touch inventory_draft.txt` and then wishes to move and rename it to `reports/system_inventory.txt`. Which core command executes this operation in place?

A. cp inventory_draft.txt reports/system_inventory.txt

B. mv inventory_draft.txt reports/system_inventory.txt

C. rm inventory_draft.txt reports/system_inventory.txt

D. touch reports/system_inventory.txt inventory_draft.txt

> **Answer: B**
>
> *The mv utility moves or renames files and directories in place from a source location to a destination.*

14. To inspect exactly the first 5 entries of /etc/passwd without opening an interactive editor, which command should you run?

A. head -n 5 /etc/passwd

B. tail -n 5 /etc/passwd

C. cat -n 5 /etc/passwd

D. less -n 5 /etc/passwd

> **Answer: A**
>
> *The head utility with `-n 5` outputs specifically the first five lines of the designated input file.*

15. An administrator wants to extract lines containing '/bin/bash' from /etc/passwd into bash_users.txt, and then append an audit comment to the end of that same file. Which sequence of operators accomplishes this?

A. Use `>` to extract the lines, then use `>>` to add the comment.

B. Use `>>` to extract the lines, then use `>` to add the comment.

C. Use `|` to extract the lines, then use `>` to add the comment.

D. Use `<` to extract the lines, then use `|` to add the comment.

Answer: A

The single `>` operator creates or overwrites the file with initial grep output, while `>>` appends additional lines without overwriting existing content.

16. Which command pipeline counts the total number of user accounts assigned the '/usr/sbin/nologin' shell in /etc/passwd?

A. grep "/usr/sbin/nologin" /etc/passwd > wc -l

B. grep "/usr/sbin/nologin" /etc/passwd | wc -l

C. cat /etc/passwd | grep -c | wc -l

D. wc -l < grep "/usr/sbin/nologin" /etc/passwd

Answer: B

The pipeline operator (`|`) routes stdout from grep directly into stdin of `wc -l`, which calculates the line count of matches.

17. While appending summary records to an audit log, a junior administrator accidentally runs `echo "Demarcation" > audit.log` instead of using `>>`. What is the result, and how should it be avoided?

A. The command errors out because `>` requires an existing file descriptor; no data is lost.

B. The existing contents of audit.log are truncated and overwritten; `>>` must be used to preserve existing log entries.

C. The echo string is appended to the beginning of the file, shifting previous records downward.

D. The operating system creates a backup copy of audit.log automatically before truncating.

Answer: B

The `>` operator truncates and overwrites target files. Appending without data loss requires the `>>` redirection operator.

18. While viewing configuration documentation in a terminal, the terminal window enters a `less` pager session and will not accept normal command prompts. How should the administrator return to Bash?

A. Press `Ctrl + C` repeatedly until the terminal aborts.

B. Type `exit` and press Enter.

C. Press `q` on the keyboard.

D. Close the terminal emulator window and start a new session.

Answer: C

When terminal output is captured inside a pager utility such as `less`, pressing `q` exits the pager cleanly back to the command prompt.

19. An administrator provisions a new local account named `developer1`. Which command ensures that the user's home directory is created and default skeleton files are copied?

A. sudo useradd developer1

B. sudo useradd -m developer1

C. sudo usermod -m developer1

D. sudo groupadd -m developer1

Answer: B

The `useradd` command requires the `-m` (make home) flag to provision a dedicated home directory for a newly created user account.

20. A user has read (`r`) and write (`w`) permissions on a directory, but lacks execute (`x`) permission. What action is the user prevented from performing?

A. Listing directory contents with ls

B. Changing their current working directory into that directory with cd

C. Viewing file metadata such as timestamps

D. Deleting files that belong to other users

Answer: B

On directories, the execute (`x`) permission is required for directory search and path traversal, meaning users cannot `cd` into a directory lacking execute permission.